

Tuesday Reading Assessment: Chapter 2-1 through 2-7

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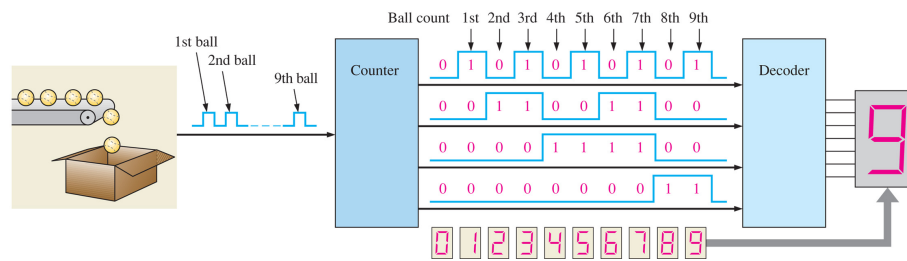


TABLE 2-2

Binary weights.

Positive Powers of Two (Whole Numbers)									Negative Powers of Two (Fractional Number)					
2^8	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	2^{-1}	2^{-2}	2^{-3}	2^{-4}	2^{-5}	2^{-6}
256	128	64	32	16	8	4	2	1	1/2	1/4	1/8	1/16	1/32	1/64
									0.5	0.25	0.125	0.0625	0.03125	0.015625

Figure 1: (Top) A sensor provides digital pulses to a counter with four outputs. (Bottom) A conversion table for binary weights.

1 Binary to Decimal, Decimal to Binary, 2's complement, Addition

- Considering Fig. 1 (bottom), convert the following binary numbers to decimal:
 - 1000
 - 1010
 - 1011
 - 1111
- Using repeated division by 2 method, convert the following decimal numbers to binary:
 - 255
 - 44
 - 64
 - 31
- What is the 2's complement of 125 in binary?
- What is $1101 + 1011$, assuming these are binary numbers?
- If the counter in Fig. 1 (top) takes 18 seconds to reach 1001, how many seconds per count?